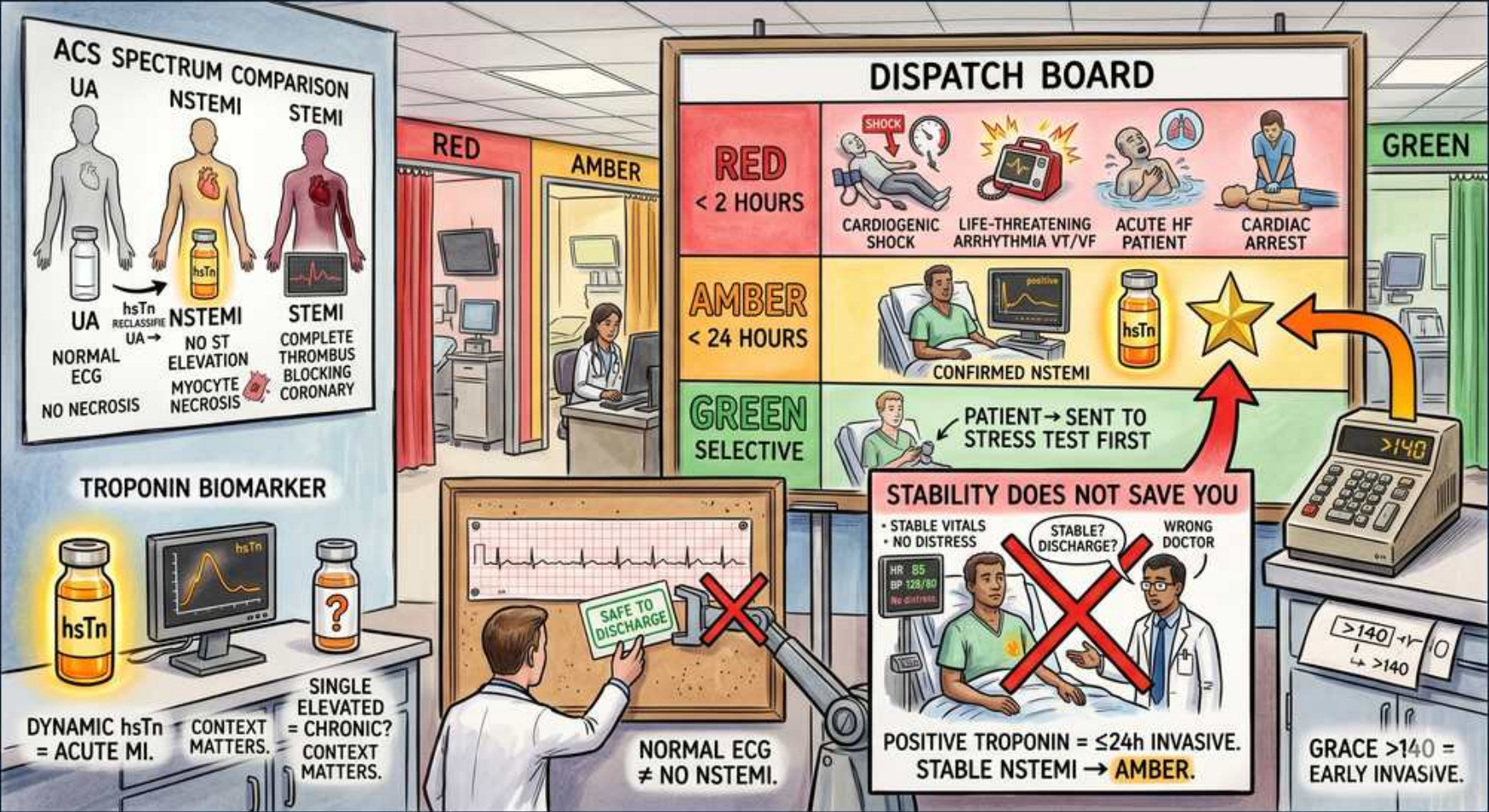


NSTEMI — ACS Spectrum, Troponin & Invasive Timing



1. UA vs NSTEMI vs STEMI

WHAT YOU SEE

'ACS SPECTRUM COMPARISON — UA / NSTEMI / STEMI'

WHAT IT ENCODES

UA = ischemic symptoms, normal troponin, no necrosis. NSTEMI = ST depression/T-inversion + troponin rise, subendocardial necrosis, non-occlusive/partial thrombus. STEMI = complete occlusion, transmural necrosis, ST elevation.

BOARD TRAP

Calling troponin-negative rest angina an NSTEMI — without troponin rise it's unstable angina.

2. UA — normal ECG, no necrosis

WHAT YOU SEE

'UA — NORMAL ECG, NO NECROSIS'

WHAT IT ENCODES

Unstable angina: rest/crescendo angina with NORMAL troponin and often non-specific ECG. With high-sensitivity troponin assays, true UA is becoming rare as small infarcts get reclassified NSTEMI.

BOARD TRAP

A normal initial troponin does not exclude evolving NSTEMI — serial high-sensitivity troponins are required.

3. STEMI — complete occlusion

WHAT YOU SEE

'STEMI — COMPLETE THROMBUS BLOCKING CORONARY'

WHAT IT ENCODES

STEMI = fully occlusive thrombus → transmural injury → ST elevation; needs immediate reperfusion. NSTEMI = subtotal occlusion managed with anticoagulation + early invasive (not lytics).

BOARD TRAP

Giving fibrinolytics for NSTEMI — thrombolysis benefits only ST-elevation (fully occlusive) MI.

4. Dynamic hs-Tn = acute MI

WHAT YOU SEE

'DYNAMIC hsTn = ACUTE MI ... SINGLE ELEVATED = CHRONIC? CONTEXT MATTERS'

WHAT IT ENCODES

A rising/falling (dynamic) high-sensitivity troponin with ischemic context = acute MI. A single stable elevation can be chronic (CKD, HF, myocarditis). Type 1 (plaque rupture) vs Type 2 (supply-demand) distinction matters.

BOARD TRAP

Diagnosing MI from one elevated troponin in CKD/HF — you need a dynamic rise/fall plus ischemic context (Type 2 vs Type 1).

5. Normal ECG ≠ no NSTEMI

WHAT YOU SEE

'NORMAL ECG ≠ NSTEMI' (man with 'SAFE TO DISCHARGE' card)

WHAT IT ENCODES

A normal or non-diagnostic ECG does NOT rule out NSTEMI — up to a third of NSTEMIs have non-specific ECGs. Diagnosis rests on troponin dynamics, not the ECG alone.

BOARD TRAP

Discharging chest pain because the ECG is normal — NSTEMI can present with a normal ECG; rely on serial troponins.

6. RED — <2 hours, immediate invasive

WHAT YOU SEE

'RED <2 HOURS — cardiogenic shock, life-threatening arrhythmia VT/VF, acute HF, cardiac arrest'

WHAT IT ENCODES

Very-high-risk NSTEMI (hemodynamic instability, cardiogenic shock, refractory angina, life-threatening arrhythmia, mechanical complication, acute HF) → IMMEDIATE invasive angiography <2 h.

BOARD TRAP

Delaying cath in an unstable NSTEMI with shock/VT — these are very-high-risk and need angiography within 2 hours.

7. AMBER — <24 hours

WHAT YOU SEE

'AMBER <24 HOURS — confirmed NSTEMI'

WHAT IT ENCODES

High-risk NSTEMI (positive troponin, dynamic ST changes, or GRACE >140) → early invasive within 24 hours. This is the standard pathway for a confirmed, stable NSTEMI.

BOARD TRAP

Treating a confirmed NSTEMI as routine — high-risk features mandate angiography within 24 h, not elective timing.

8. GREEN — selective/stress first

WHAT YOU SEE

'GREEN SELECTIVE — patient sent to stress test first'

WHAT IT ENCODES

Low-risk patients (negative serial troponins, no dynamic ECG changes, low GRACE) get a selective/ischemia-guided strategy — non-invasive stress testing before deciding on angiography.

BOARD TRAP

Rushing a low-risk, troponin-negative patient to cath — a selective ischemia-guided strategy with stress testing is appropriate.

9. Positive troponin → ≤24h invasive

WHAT YOU SEE

'POSITIVE TROPONIN = ≤24h INVASIVE. STABLE NSTEMI → AMBER.'

WHAT IT ENCODES

A positive troponin (confirmed NSTEMI) places even a hemodynamically stable patient in the early-invasive (<24 h) category. Stability does not downgrade a true NSTEMI to low-risk.

BOARD TRAP

'Stable NSTEMI' is still high-risk — positive troponin earns early invasive within 24 h.

10. Stability does not save you

WHAT YOU SEE

Banner 'STABILITY DOES NOT SAVE YOU ... STABLE DISCHARGE = WRONG DOCTOR'

WHAT IT ENCODES

Hemodynamic stability does not equal low risk in NSTEMI; risk is defined by troponin, ECG dynamics, and GRACE — not by how well the patient looks at presentation.

BOARD TRAP

Discharging a stable-appearing NSTEMI — risk stratification, not appearance, drives disposition.

11. GRACE >140 → early invasive

WHAT YOU SEE

'GRACE >140 → EARLY INVASIVE' (>140 register)

WHAT IT ENCODES

GRACE score >140 marks high-risk NSTEMI warranting early invasive (<24 h). GRACE integrates age, HR, BP, creatinine, Killip class, arrest, ST changes, and troponin.

BOARD TRAP

Ignoring GRACE — a high GRACE upgrades a troponin-positive NSTEMI to the early-invasive pathway.

12. hs-Tn biomarker — context matters

WHAT YOU SEE

'TROPONIN BIOMARKER ... CONTEXT MATTERS'

WHAT IT ENCODES

High-sensitivity troponin enables rapid 0/1-h or 0/2-h rule-in/rule-out algorithms. Interpret against the clinical context — small elevations need a dynamic change to confirm acute MI.

BOARD TRAP

Using hs-troponin as a pure on/off test — values must be interpreted with serial change and pre-test probability.
